

**Full title: Friendliness and unfriendliness are expressed sound
symbolically by the same sounds in the names of English,
Japanese, Chinese, Korean, German, and French Pokémon**

**Short title: Sound symbolic expressions of friendliness and
unfriendliness in Pokémon names**

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1. Abstract

This paper presents a cross-linguistic study of sound symbolism, analysing a six-language corpus of all Pokémon names available as of April 2021. It tests the effects of bilabial consonants and voiced plosives on a Pokémon attribute known as *friendship*. Friendship is a mechanic in the Pokémon video games that reflects how friendly each Pokémon is. This study explores sound symbolic relationships between the names of Pokémon and their default level of friendship. Many of the previously known cases of cross-linguistic sound symbolic patterns can be explained by the relationship between how sounds in words are articulated and the physical qualities of the entities that those words represent. This study, however, builds upon the underexplored relationship between sound symbolism and abstract qualities. It shows that bilabial plosives typically represent friendliness in Pokémon names while /m/, /d/, and /g/ typically represent unfriendliness. This is an important finding because cross-linguistic sound symbolic patterns that reflect abstract qualities cannot be explained by shared articulatory physiology and a natural relationship between shapes and sounds.

2. Introduction

There has been a growing interest in the exploration of sound symbolism in natural language. Studies that explore this subject typically find that the physical properties of entities are reflected in the speech sounds of their names [e.g., 1,2,3,4,5,6,7,8]. These findings challenge the first principle of languages [9] which assumes that the relationship between sound and meaning is arbitrary. For example, Sapir's [10] seminal study on sound symbolism showed that English speakers tend to associate /a/ with larger objects and /i/ with smaller objects. Arguably, Sapir's findings reflect a physical, sensory-based

relationship between the object and the sound; non-high vowels reflect largeness because the oral aperture must be larger to articulate non-high vowels [6]. This report contributes to the growing literature on the relationships between abstract qualities and sound symbolism. It examines sound symbolic relationships in the names of English, Japanese, Mandarin Chinese (hereafter: Chinese), South Korean (hereafter: Korean), German, and French Pokémon. More specifically, it examines a Pokémon attribute known as *friendship* in English and *natsuki-do* [degree of attachment] in Japanese, demonstrating that certain sounds symbolize friendliness and unfriendliness cross-linguistically.

2.1 *Sound symbolism*

Since Sapir's [10] seminal study, quantitative research has revealed many sound symbolic patterns, several of which hold cross-linguistically. For example, voiced obstruents have been found to be judged to be more suitable for the names of larger or heavier objects [11]. This pattern has been observed in English [e.g., 11], Japanese [6,12,13], and Chinese [6]. Alongside size, shape has also been found to be reflected sound symbolically. Köhler [5] was the first to notice that angular objects tend to be associated with words like *takete* while round objects tend to be associated with words like *maluma*, the effect subsequently replicated with *kiki* and *bouba*, respectively [14]. These sound-meaning associations are known to hold across different languages [15,16,17, but see 18,19]. It is yet unclear, whether consonants or vowels play the main role in the bouba/kiki effect, or whether it relies on their additive symbolic relation [20,21,22]. However, the findings so far signify that this particular shape-sound relationship is relevant for the acquisition and development of language [23].

Several explanations have been presented to explain the relationship between sounds and physical properties. Regarding the size–sound relationship, Ohala [24,25] proposed the underlying mechanism to be the “frequency code”. In the natural environment, animals which are larger naturally have a lower fundamental frequency (F0). In turn, lower F0 is perceived as a signal of largeness, and therefore threat and dominance, while higher F0 as a signal of smallness that stands for the lack of threat and friendliness. It has been shown that some animals use this correspondence as a deceiving mechanism in order to signal hostility [26,27]. Ohala [24,25] based his frequency code on F0, because different vowels have different intrinsic F0: e.g., [a] has a lower F0 than [i] [28]. He, however, also refers to the second formant frequency (F2) as a possible cause for this effect in sound symbolism [25]. F2 frequency has been found to iconically express deictic relations [29,30,31,32] such that a proximal reference is conveyed by vowels with higher F2 and distal by vowels with low F2. Johansson & Zlatev [29], after Traunmüller [30], propose that the proximal–distal and small–large oppositions are based on the iconic similarity between mappings. A small size corresponds to a small distance, and a large size to a large distance [29].

Thus, the correspondences described above rely on iconicity – a relationship between the linguistic form and meaning [e.g., 33]. They are not restricted to speech sounds, but extend to other linguistic domains, such as prosody. It has been shown that listeners infer information regarding size [34] or movement [35] of an entity from prosodic information. Higher pitch represents smaller entities, and faster speech rate represents entities in faster motion. These mechanisms are also conveyed in speech production. It was shown that speakers reliably use higher pitch to convey “smaller” domains [36, 37], or exhibit higher

speech rate when talking about fast events [37,38,39]. Some aspects of iconic prosody can be transferred to the written domain. Fuchs, Savin, Solt, Ebert & Krifka [40] investigated adjective antonym pairs, such as short/long, tiny/huge, and fast/slow, in an English blogger corpus. They found that the adjectives belonging to an iconically “larger” domain, do exhibit letter replications more often than the adjectives that belong to the “smaller” domain. Therefore, it seems that iconically the amount of the meaning can be expressed with the amount of sound or graphemes, respectively.

Cross-linguistic iconic coding principles may be natural tendencies to express meanings with sounds that correspond to the physical nature of entities. Human languages are produced by a shared physiology, so it makes sense that we find cross-linguistic patterns whereby large objects are represented by large (or more) sounds and round objects are represented by sounds that are produced with lip rounding gesture. While size and shape can be expressed physically, more recent studies in sound symbolism have explored possible relationships between speech sounds and more abstract qualities such as humour [41,42] and rudeness [43]. For example, studies have shown that humans languages signal emotion via sound symbolism. Adelman, Estes, and Cossu [44] examined word initial phonemes in English, Spanish, Dutch, German, and Polish and found that across languages, the first phoneme in each word is a significant predictor of emotional valence. They also found that phonemes that were uttered faster tended to convey danger, proposing that emotional sound symbolism is an evolutionary adaptation in humans akin to alarm calls found in animal communication systems. Another example is that in Japanese, voiced obstruents tend to be associated with negative images [e.g., 12] while bilabial consonants are often associated with infancy [45,46]. The latter finding is

likely grounded in speech development. Babies use their lips to suck milk, and as such they are the most relevant organ for them: as a result, labials are among the very first consonants they produce [47]. Although infancy is expressed physically, this finding motivated a study by Uno et al., [48] which explored sound symbolism in names of villainous and non-villainous characters. They used a corpus of the names of English Disney characters and the names of Japanese Pokémon. Pokémon have associated “types” and they used “fairy type” as a proxy for non-villainous and “dark type” (in Japanese *aku-taipu* [evil type]) as a proxy for villainous. In a similar analysis as is found in the present study, they found that across both languages, voiced obstruents were associated with villainous names and bilabial consonants were associated with non-villainous names. The present study expands this finding by testing for a relationship between friendliness and these speech sounds in the names of Pokémon.

2.2 *Pokémonastics*

Pokémon—also known as *Pocket Monsters* in Japan—is a Japanese media franchise cantered on fictional monsters called Pokémon. Human characters in the franchise—known as trainers—catch and train Pokémon to battle with other Pokémon trainers for sport. The franchise began as a video game which soon developed into a mixed media franchise. In the video game world, Pokémon have attributes which are expressed as numerical values. Pokémonastics is a relatively new field of research that examines sound symbolic relationships between the names of Pokémon and these attributes. Initially, these studies focused on Japanese Pokémon names and found that features such as number of voiced obstruents and name length reflected Pokémon attributes such as increased size and strength parameters [4]. These studies have since been expanded to examine

Pokémon names cross-linguistically to determine whether patterns exist across languages. In one such study, Shih, Ackerman, Hermalin, Inkelas & Kavitskaya [49] found that increased name length represented increased size, power, and evolution level not only in Japanese but also in English Pokémon. As Pokémon battle, they gain experience, once they have gained enough experience, their level and attributes increase. Most Pokémon have the option to evolve once they have attained a certain level which typically results in increased size and combat ability. Given that size, power, and evolution level are related, this finding can arguably be attributed to *iconicity of quantity* [50]—whereby larger entities are expressed by longer words as both strength and size [49] and evolution level can be symbolically expressed in terms of the length of Pokémon names [51].

An interesting hypothesis that has arisen from Pokémonistic studies is that elements of sound symbolism in human languages may reflect attributes that are crucial to evolutionary survival [49]. Pokémon attributes that reflect survivability—such as size and combat parameters—seem to be robustly reflected sound symbolically in Pokémon names while parameters that have little effect on survivability—such as gender—are not. Although not explicitly stated by Shih et al., [49], this suggests that elements of sound symbolism relating to survivability can be innate [52,53]. This proposal raises the possibility that human language evolved—at least somewhat—in the oral-aural modality rather than purely kinesic (arguments for/against this proposal are presented in [54] and [55] respectively). While much more evidence is required to test this hypothesis, if it is on the right track, it will have major implications for our understanding of the origins of human language. This is one of the reasons why it is important to explore sound symbolic patterns that reflect abstract qualities because cross-linguistic patterns that reflect

physicality can be explained by natural and iconic relationships between sounds and objects. When a quality has no physical representation, this explanation cannot hold [56]. In order to contribute to addressing this general question, the present study seeks to test sound symbolic relationships pertaining to friendliness, with a caveat in mind that whether or not friendliness is an abstract quality is debatable given that friendliness may be interpreted as a lack of threat or a lack of dominance which are conceptualized as concrete qualities in the frequency code hypothesis [56].

2.3 *Specific hypotheses tested*

In the Pokémon video game series, friendship is a mechanic that is largely hidden from the player. Friendship is a metric that reflects how much the Pokémon likes its trainer (expressed as a value ranging from 0-255). If a trainer treats their Pokémon well (such as healing it during battle) the friendship value increases, if they treat their Pokémon poorly (such as making it battle until its hit points are exhausted), the value decreases. Each Pokémon has a default level of friendship, in the following we refer to this value as *base friendship*. Base friendship is the friendship value that each Pokémon has when it is first encountered by the player. Pokémon can be traded between players and when a Pokémon is traded all its attributes remain the same except for its friendship attribute which reverts to base friendship. Base friendship may therefore be considered a measure of how naturally friendly each Pokémon is. The present study uses the base friendship parameter as a predictor for the frequency of bilabial consonants and voiced obstruents. It tests the following hypotheses:

1. Voiced obstruents will be associated with unfriendly Pokémon.
2. Bilabial consonants will be associated with friendly Pokémon.

173 3. Given that /b/ is both a voiced obstruent and a bilabial consonant, we make no
174 predictions as to its association with friendliness.

175

3. Materials and Methods

The following analysis is conducted on Pokédex tables extracted from Bulbapedia (<https://bulbapedia.bulbagarden.net>, last access April 2021). Bulbapedia provides names for Pokémon in English, Japanese, Chinese, Korean, German, French, Thai, and Russian. Thai and Russian names are not included in the following analysis because these are directly assimilated from Japanese and English names, respectively. This analysis examines the names of all Pokémon that are currently featured in the video game series (as of April 2021). There are a total of 898 Pokémon spanning generations 1 through 8 (1996-2021). Given that we examine six languages, the following analysis is conducted on a corpus of 5,388 Pokémon names. In the following, we examine the relationship between a statistic known as base friendship and the frequency of graphemes that represent voiced plosives and bilabial consonants. Unlike previous Pokémon studies, Poisson regression is used to predict the occurrence of certain sounds based on base friendship, because while base friendship is represented numerically, it is not a continuous variable. Table 1 reports the distribution of Pokémon per level of base friendship.

Base Friendship	Number of Pokémon
0	48
35	73
50	744
70	2
90	5
100	15
140	11

Table 1. Distribution of Pokémon per level of base friendship.

The following analysis examines graphemes from the names of English, German and French Pokémon as well as Japanese Hepburn romanization, Chinese pinyin, and Korean McCune-Reischauer romanization. It is important to note at this point that this

study on the sound symbolic nature of Pokémon names is based on a graphemic analysis of six languages because the writing systems analysed do not constitute a 1:1 match to speech sounds, particularly in the case of English, German, and French. To address this issue, we only analysed graphemes that constitute a relatively good match to the sounds they represent and omitted digraphs which generally do not represent the sounds under investigation. Future investigation in this area would ideally be conducted on a phonetic analysis but given we examined more than 5000 Pokémon names, across six languages, a phonetic analysis was not feasible for the present study. In the following we report on the frequency of five graphemes: <p>, , <m>, <d>, and <g>. Alongside the reasons noted above, these graphemes were selected based on the findings of Uno et al., [48] which showed that voiced obstruents and bilabial consonants were predictors of villainy. Other voiced obstruents were not included because they do not occur—or are not represented graphemically—in all the languages observed.

It is also important to note that a certain amount of name borrowing occurs between languages. For example, the Pokémon known as Pikachu in English (base friendship = 50), is written as <pikachū> in Japanese Hepburn romanization, <píkǎqiū> in Chinese Pinyin, <p'ik'ach'yu> in Korean McCune-Reischauer romanization, and <pikachu> in both German and French. While Pikachu is a special case because it is a mascot for the franchise, the reader should be made aware that borrowing does occur between languages and will therefore affect the upcoming results as we have not omitted any Pokémon names. Table 2 presents a list of the names in all languages of 9 Pokémon from the first generation of the game as an example. These Pokémon are known as the

217 “starter” Pokémon because they are—or evolve from—the Pokémon that the player can
 218 select from at the start of the first generation of the games.

English	Japanese	Chinese	Korean	German	French
Bulbasaur	Fushigidane	Miàowāzhǒngzǐ	Isanghaessi	Bisasam	Bulbizarre
Ivysaur	Fushigisō	Miàowācǎo	Isanghaep'ul	Bisaknosp	Herbizarre
Venusaur	Fushigibana	Miàowāhuā	Isanghaekkot	Bisaflor	Florizarre
Charmander	Hitokage	Xiǎohuǒlóng	P'airi	Glumanda	Salamèche
Charmeleon	Rizādo	Huǒkǒnglóng	Rijadū	Glutexo	Reptincel
Charizard	Rizādon	Pēnhuǒlóng	Rijamong	Glurak	Dracaufeu
Squirtle	Zenigame	Jiéníguī	Kkobugi	Schiggy	Carapuce
Wartortle	Kamēru	Kǎmīguī	Ōnibugi	Schillok	Carabaffe
Blastoise	Kamekkusu	Shuǐjiànguī	Kōbugwang	Turtok	Tortank

219 **Table 2:** List of each of the starter Pokémon from Generation 1 in each language.

220
 221 To ensure that the graphemes we were examining were a good match to the speech
 222 sounds they represent, certain digraphs were omitted from the following analyses.
 223 Instances of doubling (e.g., <pp>) were counted as single instances of each sound.
 224 Doubling represents gemination in Japanese Hepburn romanization and Korean McCune-
 225 Reischauer romanization, and is typically produced as singleton in English, German, and
 226 French. Geminate consonants were counted as single occurrences—rather than double—
 227 because while closure duration is longer for geminates, articulatorily speaking, geminates
 228 consist of one (longer) closure event followed by one release event, just like
 229 corresponding singleton consonants. Sequences of <ph> were omitted because these
 230 typically represent [f] in English, German, and French and do not occur in other languages
 231 in the corpus. Sequences of <pf> were omitted because these represent the German
 232 affricate (cf. ‘Pferd’ [pʰɛʁt] *horse*). <pf> does not occur in other languages in the corpus.
 233 There were four instances of the digraph <dg> that were omitted. These all occurred in
 234 the English corpus and corresponded to [dʒ] (e.g., *Pidgey* [pɪdʒi:]). <gn> digraphs were

excluded from the French analysis as this digraph typically represents [ɲ] (e.g., champagne [ʃɑ̃paɲ]). Lastly, <ng> digraphs were omitted from the analysis in all languages except for Japanese (where the [g] is predictably produced). This is problematic because <ng> can represent either [ŋ] or [ŋɡ] in English (e.g., the <g> in *singer* is silent while the <g> in *finger* is not), and can represent either [Ṽg], or [Ṽʒ] in French (e.g., *anglias* [ãɡlɛ], *congé* [kɔ̃ʒe]). While pronunciation rules do exist for English and French, they are not always adhered to. For the sake of simplicity, <ng> was omitted from the English and French analyses. The <ng> digraph typically represents [ŋ] in Chinese Pinyin, McCune-Reischauer romanization in Korean, and in German. Table 3 provides a count of the instances of each grapheme and the omitted digraphs.

Count	<p>		<m>	<d>	<g>	all
	1148	962	1550	1112	1835	6607
Omit	<pp>	<bb>	<mm>	<dd>	<gg>	
	75	20	21	10	55	
	<pf>			<dg>	<dg>	
	7			4	4	
	<ph>				<gn>	
	38				31	
					<ng>	
					783	
Total Omit	120	20	21	14	873	1017
Analysis Totals	1028	942	1529	1098	962	5590

Table 3. Count of the instances of each grapheme and the omitted digraphs.

4. Results

A descriptive analysis shows that there is a relationship between base friendship and the frequency of the five graphemes, whereby <p> occurs most frequently in Pokémon with a base friendship of 140, while <g> occurs most frequently in Pokémon with a base friendship of 0. Table 4 and Figure 1 report the average number of times that each grapheme occurs in each Pokémon name distributed by base friendship level.

Base Friendship	0	35	50	70	90	100	140
<p>	0.08	0.13	0.20	0.17	0.03	0.22	0.45
	0.07	0.15	0.18	0.33	0.10	0.19	0.09
<d>	0.38	0.24	0.29	0.33	0.07	0.24	0.20
<m>	0.21	0.24	0.20	0.25	0.37	0.10	0.17
<g>	0.30	0.19	0.17	0.08	0.03	0.07	0.05

Table 4: Average number of times that each grapheme occurs in Pokémon names by base friendship level.

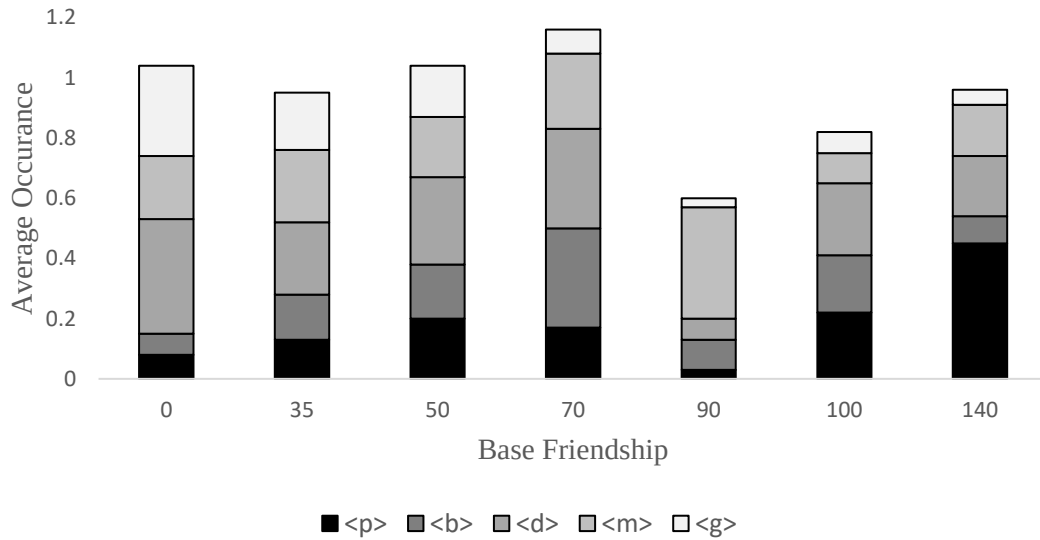


Figure 1: Stacked bar plot illustrating the relationship between average occurrence of each grapheme and base friendship level.

Since we are dealing with count data (e.g. “how many times does [p] occur?”), a series of Poisson regression analyses were conducted with base friendship as a predictor

for each of the five graphemes (for a clear exposition for this statistical analysis, see [57]: Ch 13). Language was included as a random variable in this analysis as it was conducted on the entire dataset (results reported in Table 5). These results show a significant effect ($p < 0.001$ in all cases) of base friendship on grapheme frequency where graphemes that represent bilabial plosives (/p/ and /b/) are associated with high base friendship while <d>, <m>, and <g> graphemes are associated with low base friendship.

<i>Preds.</i>	<i>R²</i>	<i>z-value</i>	<i>p</i>
<p>	0.044	15.77	< 0.001
	0.006	5.49	< 0.001
<d>	0.004	-4.37	< 0.001
<m>	0.011	-7.69	< 0.001
<g>	0.041	-14.9	< 0.001

Table 5: Results of a series of Poisson regression analyses conducted to test base friendship as a predictor for grapheme frequency in all languages.

These associations generally hold true when each language is analysed individually. Table 5 reports on a series of Poisson regression analyses conducted on each language separately.

	<i>Preds.</i>	<i>R</i> ²	<i>z-value</i>	<i>p</i>
English	<p>	0.019	4.16	< 0.001
		0.009	2.93	0.003
	<d>	0.002	-1.22	0.221
	<m>	0.004	-1.98	0.048
	<g>	0.089	-8.97	< 0.001
Japanese	<p>	0.118	10.86	< 0.001
		0.003	1.65	0.098
	<d>	0.023	-4.57	< 0.001
	<m>	0.011	-3.08	0.002
	<g>	0.029	-5.07	< 0.001
Chinese	<p>	0.038	6.01	< 0.001
		0.009	2.8	0.005
	<d>	0.002	1.31	0.192
	<m>	0.008	-2.67	0.008
	<g>	< 0.001	0.44	0.66
Korean	<p>	0.061	7.6	< 0.001
		0.001	0.96	0.335
	<d>	0.011	-3.17	0.002
	<m>	0.021	-4.35	< 0.001
	<g>	0.039	-5.93	< 0.001
German	<p>	0.054	7.16	< 0.001
		0.024	4.69	< 0.001
	<d>	0.022	-4.44	< 0.001
	<m>	0.02	-4.26	< 0.001
	<g>	0.105	-9.78	< 0.001
French	<p>	0.019	4.17	< 0.001
		< 0.001	0.56	0.578
	<d>	0.001	0.69	0.49
	<m>	0.007	-2.49	0.013
	<g>	0.05	-6.68	< 0.001

Table 5: Results of a series of Poisson regression analyses conducted to test base friendship as a predictor for grapheme frequency in each language separately.

In all languages, base friendship is a significant predictor of how often <p> occurs ($p < 0.001$ in all cases), whereby the names of Pokémon with high base friendship are more likely to be represented with <p>. High base friendship trends towards increased frequency of in all languages but this is only significant in English ($p = 0.003$),

Chinese ($p = 0.005$), and German ($p < 0.001$). The relationship between base friendship and <d> frequency varies between languages. In Chinese and French there is a small but insignificant positive correlation between base friendship and <d> frequency. In all other languages except for English, low base friendship is found to be a significant predictor of <d> frequency. Base friendship is a significant predictor of <m> frequency in all languages ($p < 0.05$ in all cases), whereby low base friendship is associated with increased <m> frequency. Interestingly, base friendship is negatively correlated with <g> frequency in all languages except Chinese. In all other languages, base friendship is a robust and significant predictor of <g> frequency ($p < 0.001$).

A potential concern that can be raised against this conclusion may be that these results mirror those findings in earlier studies that have shown that voiced obstruents are related to increased size[e.g., 4,45] while bilabial consonants are related to cuteness in English and Japanese [45] and possibly other languages. To examine a potential relationship between base friendship, size (height and weight), and the sum of the combat parameters (attributes known as hit points, attack, defence, special attack, special defence, and speed), we conducted a multivariate regression analysis to explore base friendship as a predictor of combat and size measures. As was observed in Kawahara, Noto & Kumagai [4], Pokémon height and weight measures are heavily right-skewed, so we took the natural logarithm of these measures. The results show that base friendship is negatively correlated with weight, and the total of the combat attributes. The results of this analysis are presented in Table 6.

Predictors	Estimates	t-value	p
combat_total	0.06	7.23	<0.001
weight_log	21.74	9.58	<0.001
height_log	3.26	0.52	0.604
combat_total * weight_log	-0.05	-10.4	<0.001
combat_total * height_log	-0.03	-2.04	0.042
weight_log * height_log	32.94	12.05	<0.001
(combat_total * weight_log) * height_log	-0.06	-10.51	<0.001

Table 6: Results of a multivariate regression analysis examining the relationship between base friendship and combat and size attributes.

This is thus a legitimate concern given that friendly Pokémon are indeed smaller and weaker than unfriendly Pokémon. We conducted a series of multivariate Poisson regression analyses examining base friendship, size, and combat variables as predictors of the frequency of each grapheme to determine which variable was the better predictor of grapheme frequency. These are reported in Table 7. The results of the series of multivariate Poisson regression analyses show that base friendship is a better predictor of <p> and <g> frequency than size and combat metrics, but not of , <d>, and <m> frequency.

Grapheme Preds.	<p>				<d>		<m>		<g>	
	z-value	p	z-value	p	z-value	p	z-value	p	z-value	p
B. Friendship	5.42	<0.001	1.59	0.112	-0.06	0.951	-4.32	<0.001	-3.96	<0.001
Combat Total	-2.58	0.01	-3.22	0.001	-0.63	0.529	2.43	0.015	0.96	0.337
Log Height	0.51	0.61	0.01	0.993	3.22	0.001	-5.44	<0.001	-0.07	0.943
Log Weight	-2.2	0.028	0.15	0.884	0.21	0.833	0.83	0.408	2.64	0.008
	R ² = 0.25		R ² = 0.009		R ² = 0.017		R ² = 0.019		R ² = 0.024	

Table 7: Results of a series of multivariate Poisson regression analyses examining base friendship, size, and combat variables as predictors of the frequency of each grapheme in all languages.

5. Discussion

The current study contributes to the body of literature that examine the relationships between sounds and meanings in natural languages. It examines a corpus of over 5,000 Pokémon names from six languages and shows a clear relationship between the Pokémon attribute known as base friendship and the graphemes that represent sounds included in the names of Pokémon. We hypothesized that bilabial consonants would occur more frequently in the names of friendly Pokémon while unfriendly Pokémon names would contain voiced obstruents. The findings are partly in line with the hypotheses, as bilabial segments in general were found to be correlated positively with base friendship. It is, however, the case, only for the plosives /b/ and /p/, not for the nasal bilabial /m/. On the contrary, the nasal bilabial /m/ signifies unfriendliness in our data. This is a novel result in light of Kumagai [45] and Kumagai & Kawahara's [46] results, who found a positive correlation between bilabial segments and infancy. Considering Ohala's [25] frequency code, infancy is a signal for lack of threat. Infants famously evoke infant-directed speech in adults and prefer this over adult-directed speech [58]. Our analysis is the first one that disentangles different bilabial segments and exposes this contradicting effect when considering base friendship rating in Pokémon. Indeed, <m> and <g> were shown to be significant predictors of unfriendliness. The one notable exception being that the relationship between unfriendliness and <g> frequency in Chinese did not adhere to the pattern exhibited by other languages.

Base friendship is arguably an abstract quality, it should have no physical representation. It has, however, been shown that seemingly abstract forms may evoke perceptual, affective, and sensory response similar to real-world objects. In Lindauer's [59] experiment, participants judged the shapes of *takete* and *maluma*, originally

conceived by Köhler [5], as being the least and the most friendly, respectively, as compared to neutral stimuli. In another experiment, Sidhu, Deschamps, Bourdage & Pexman [60], examined how sound symbolism conveys abstract qualities. The authors investigated the perception of various personality traits in first names. The names were grouped into those that contained sonorants /m/, /n/, and /l/, and those that contained voiceless stops /p/, /t/, and /k/. The results suggest that sonorants and voiceless stops are associated with different personality factors, some of which may be linked to friendliness [60].

In the current study we report findings that friendliness may also have a physical representation, as friendly Pokémon tend to be smaller and weaker than their unfriendly counterparts. An attentive reader may nevertheless ask, whether the current study tests friendliness explicitly, or whether our results are simply reflecting known size effects in sound symbolism. The multivariate analysis that was conducted to test this possibility did show that base friendship was a better predictor than combat, weight, and height parameters for <p> and <g> frequency, but not for other graphemes. Voiced obstruents have been shown to be predictors of size in other Pokémonistic studies. Although it was not found to be a significant predictor in Japanese, Korean, and French, high frequency trended toward high friendliness in all languages, suggesting that friendliness and size are expressed differently sound symbolically. The relationship between the frequency of the bilabial nasal grapheme <m> and base friendship presents another piece of evidence that base friendship is not simply a reflection of size. Base friendship was shown to be a significant predictor of <m> frequency in all languages, whereby Pokémon with low base friendship generally had a greater occurrence of <m> in their names. This

is a surprising finding given that bilabial consonants (in English and Japanese at least) appear to be associated with cuteness and infancy [45]. Important here is the relationship between size parameters and <m> frequency. Weight was not shown to be a significant predictor of <m> frequency, but height was shown to connote shortness (although this was only shown to be significant in Japanese and Chinese). Given the negative relationship between base friendship and <m> frequency, if base friendship is simply a reflection of Pokémon size, then a positive relationship between height and <m> frequency would be expected. This was clearly not the case.

It is interesting to examine our findings through the lens of Shih et al.,’s [49] hypothesis that elements of sound symbolism possibly reflect evolutionary survivability. Here however, we divert from their assumption that symbolism in human languages communicates the qualities of one’s allies and tentatively put forth our own hypothesis that relates more closely to Adelman, Estes, and Cossu’s [44] alarm call theory. Pokémon are both allies and enemies in the Pokémon world. In the non-Pokémon world, many non-human primates have elaborate alarm calls that communicate information regarding approaching predators. East African Vervet Monkeys, for instance, have an alarm call system that has evolved to differentiate between mammalian, avian, and snake shaped predators [61]. It is possible therefore that elements of sound symbolism in human languages is the result of a vestigial alarm call system which early humans used to warn others of approaching danger as hypothesised by Adelman, Estes, and Cossu [44]. In the context of the findings of the present study, perhaps patterns relating to base friendship are the result of an alarm call system that communicated the potential threat of an observed predator. This may explain the apparent sound symbolic relationship between

demeanour and size because larger predators would be more likely to engage early humans as prey. This also may give insight into why the relationship between voiced obstruent frequency and weight appears to have a greater effect size than height in Pokémon studies [4,49]. In professional combat sports, fighters are separated into weight classes—rather than height classes—because weight is a better predictor of threat. We find this proposition of a sound symbolic pattern that reflects threat more convincing than one that expresses the survivability on one's allies because of convergent evolution—that under the same environmental pressures, separate species will exhibit similar traits—and the prevalence of alarm call systems in the animal kingdom. That said, we note that the present study does not constitute sufficient evidence to substantiate this hypothesis but discuss it here with the hope that further investigation may yield interesting results.

 Ideally, this study would have been conducted on a phonetic analysis, rather than a graphemic one. A phonetic analysis would open interesting areas of enquiry into whether sound symbolism is phonetically or phonemically based. In German for instance, consonants generally undergo devoicing at the ends of words and lexical morphemes. This phenomenon was not accounted for in the present study because German Pokémon names are often compounds of both native and non-native words which may not adhere to German phonological rules. The German Pokémon name, *Dartignis* for example, is composed of the German *dart* [dart] and the Latin *ignis* [fire]. German phonological rules would dictate that the /g/ in *ignis* were devoiced but this is not the case because the production of *ignis* in the name conforms to Latin phonology. A phonetic transcription

undertaken by phoneticians with extensive knowledge of Pokémon names would therefore be needed to conduct such an analysis.

Another consideration we made in the current study was to omit names that were borrowed (as in the case of Pikachu). Borrowing can be problematic in a cross-linguistic study because it has the potential to overstate results. In practice, however, omitting names would be difficult because it is not always clear when a name is borrowed and when it comes from a similar source. For example, the Pokémon known as *Omanyte* in English resembles a species of extinct marine molluscs known as *ammonites*. *Omanyte* is known as *Omunaito* in Japanese, *Júshíshòu* in Chinese, *Amnait'ŭ* in Korean, *Amonitas* in German, and *Amonita* in French. Given that most of these names are clearly based on the scientific name for ammonite (ammonite translates into *zhànyào* in Chinese), it is difficult to ascertain whether these are true borrowings, or whether they are based on the same source.

6. Conclusion

The present study shows a clear cross-linguistic relationship between frequency of graphemes and the Pokémon attribute known as base friendship. It shows that <p> and represent friendliness in all the examined languages, while <m> and <g> are shown to represent unfriendliness in all languages except Chinese. This is an important finding because friendliness is arguably an abstract quality and can therefore not be explained by a natural relationship between the shape of an object and the shape that the oral cavity conforms to when making the sounds that describe or name said object. The finding that sound symbolic patterns pertaining to friendliness hold cross-linguistically suggests that more is at play in sound symbolism than a natural relationship between shape and sound.

429 **7. Acknowledgements**

430

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